

Business Question

If StreamHub personalizes thumbnail artwork per user, will it meaningfully improve engagement (clicks, watch time) - and just as importantly, will it help or hurt subscriber retention?

Dataset

streamhub_ab_test_data.csv - 80,075 rows, split roughly 50/50 between control and variant, tracked over a 30-day test window.

Column	Meaning
user_id	Unique user identifier
group	control or variant
region	India / USA / UK / Southeast Asia / Other
device	Mobile / Smart TV / Web / Tablet
subscription_tier	Free or Premium
browse_impressions	How many times thumbnails were shown to this user
title_clicks	How many thumbnails they clicked
total_watch_minutes	Total minutes watched over the 30 days
sessions_count	Number of distinct viewing sessions
completed_titles	Number of titles watched to completion
churned_30d	Whether they cancelled/went inactive during the window

streamhub_daily_trend.csv - daily aggregated impressions/clicks/CTR per group, for all 30 days.

Note: This data is NOT pre-cleaned. It contains missing values, duplicate rows, duplicate user IDs with conflicting data, impossible values (clicks greater than impressions, negative watch minutes), and inconsistent text formatting.

Section 0 - Data Cleaning & Validation (Basic)

- **Q1.** Load the dataset. What's the shape (rows, columns)? Are there missing values in any column, and how many?
- **Q2.** Are there exact duplicate rows? Are there duplicate user_ids with conflicting data? How will you handle each case?

- **Q3.** Are there impossible values rows where title_clicks is greater than browse_impressions, or negative total_watch_minutes? How many, and what will you do with them?
- **Q4.** Are categorical columns like region and device consistently formatted, or mixed casing/extra whitespace? Clean these to one consistent label each.
- **Q5.** After cleaning, compute click_through_rate = title_clicks / browse_impressions per user. Handle division-by-zero cases.

Section 0B - Organize, Pivot & Visualize (Basic)

- **Q6.** Build a pivot table showing average click_through_rate and average total_watch_minutes by group and device.
- **Q7.** Plot the distribution of total_watch_minutes. Is it skewed? Mean or median?
- **Q8.** Plot a bar chart comparing overall click_through_rate between control and variant.

Section 1 - Ethics Check (Intermediate)

- **Q9.** Walk through Risk, Benefit, Choice/Alternatives, Data Sensitivity for this test. Does personalizing based on viewing history raise data sensitivity concerns?

Section 2 - Choosing Metrics (Intermediate)

- **Q10.** Map the funnel from "browse page shown" to "subscriber retained." Primary metric? Secondary metric? Guardrail metric?
- **Q11.** Given the shape from Q7, should total_watch_minutes use mean, median, or a nonparametric method?

Section 3 - Experiment Design (Intermediate/Advanced)

- **Q12.** What is the unit of diversion? Is it consistent?
- **Q13.** Why might a cohort design suit churned_30d better than a population design?
- **Q14.** Calculate required sample size for a 2pp lift on ~12% baseline CTR, 95% confidence, 80% power. Is this dataset big enough?

Section 4 - Statistical Analysis (Advanced)

- **Q15.** Sanity check the group split (Sample Ratio Mismatch test).
- **Q16.** Test click-through-rate for statistical AND practical significance.
- **Q17.** Test total_watch_minutes using your justified method from Q11.
- **Q18.** Test the guardrail metric churned_30d.

Section 5 - Segments & Learning Effects (Advanced)

- **Q19.** Break CTR down by device. Consistent, or does one segment differ?
- **Q20.** Break it down by subscription_tier. Any Simpson's Paradox surprise?
- **Q21.** Plot daily CTR trend. Stable lift, or fading novelty effect?

Section 6 - Final Recommendation (Advanced)

- **Q22.** Write a 150-250 word executive summary: findings, recommendation, what to test next.

Deliverables

- Cleaned dataset with data-quality notes
- Completed notebook answering all 22 questions
- At least 5 charts
- README.md
- (Optional) Power BI or Streamlit dashboard